

AUTHENTICLENS

KNOWLEDGE BASE - DOCUMENT 3 OF 9

DOCUMENT 3 OF 9 - REALISM PROMPTING FORMULAS

Corrected prompt engineering system for Gemini, Nano Banana, Nano Banana Pro,
and identity-preserving realism edits

Version 2.0 - April 2026. Fully revised and corrected from the original v1.0 document. This version integrates the audit findings, errata, Lightroom/editorial separation, mobile/compression caveats, provenance limits, and Gemini / Nano Banana / Nano Banana Pro execution requirements.

Prepared for: AuthenticLens - The Realism Architect

Target use: Gemini Gem knowledge-base upload

Primary image workflow: Gemini Pro reasoning + Nano Banana / Nano Banana Pro image generation and editing

Language: English (Canada)

Purpose

This document is the corrected prompt-engineering reference for AuthenticLens. It updates the original realism vocabulary and scenario formulas for Gemini-style instruction following and Nano Banana / Nano Banana Pro image generation/editing. It replaces blunt universal negative prompts with preservation-first, task-specific prompt architecture.

Intended Gemini Behaviour

- Translate the user request into a precise edit plan before generating or editing.
- Preserve identity, face, body, complexion, hairstyle, expression, and clothing unless the user explicitly requests a change.
- Use Nano Banana / Nano Banana Pro with instruction-style prompts: task, preservation constraints, target edit, lighting/camera/material realism, negative constraints, and final QC.
- Avoid overprocessing. Realism comes from coherent physics, texture, contact shadows, and context, not from heavy grain or blur.
- Use the prompt modules as composable blocks, not as one massive prompt dumped into every task.

Version 2.0 Corrections Applied

Old weakness	Updated correction
Prompting focused on Midjourney/Stable Diffusion/DALL-E/Firefly.	Added Gemini / Nano Banana / Nano Banana Pro instruction architecture and multi-turn repair workflow.
Universal negative prompt was too broad.	Use targeted negative constraints based on diagnosed failure type. Avoid suppressing real details.
No identity preservation layer.	Added preservation-first rules for faces, bodies, complexion, expression, clothing, and background.
No clothing swap/product/collage/background modules.	Added prompt modules for fashion, Saint Laurent-style outfit swaps, Melato product images, group collages, and Ottawa/Rideau Street spring backgrounds.
Camera/film language could become decorative.	Use camera/film details only when they fit the image context and user goal.

Gemini / Nano Banana Naming and Use Rule

User shorthand: “Nano Banana 2 Pro” should be interpreted as a request for the best current Gemini image-generation/editing workflow. Formal documentation should say Nano Banana / Nano Banana Pro. Current Google source note: Nano Banana = Gemini 2.5 Flash Image (gemini-2.5-flash-image); Nano Banana Pro = Gemini 3 Pro Image Preview (gemini-3-pro-image-preview).

Master Prompt Architecture

Use this reusable structure for most Gemini image edits and generations. The order matters: preservation constraints must appear before stylistic or realism instructions.

Prompt segment	Purpose	Template language
1. Task command	State the edit/generation objective.	Edit the provided image to [specific outcome].
2. Preservation constraints	Lock identity and non-target elements.	Preserve the subject identity, face, age, complexion, body structure, hairstyle, expression, pose, and recognisable appearance.
3. Target change	Name exactly what may change.	Only modify [lighting/background/clothing/texture/perspective]. Do not modify [protected elements].
4. Scene/context	Ground environment and intent.	Set the scene in [location/season/time/weather] with realistic everyday details.
5. Camera/lens behaviour	Anchor optical realism.	Make it look like a real photo captured on [phone/35mm/full-frame] with [lens/DOF/exposure] appropriate to the scene.
6. Light and shadow	Force integration.	Match light direction, colour temperature, contact shadows, reflections, and ambient occlusion.
7. Material texture	Prevent plastic outputs.	Preserve natural skin texture, fabric weave, seams, stitching, lint, wear, and realistic material response.
8. Editing style	Define post-processing level.	Use a restrained Lightroom-style grade: natural contrast, accurate colour, subtle grain, no heavy HDR.
9. Negative constraints	Block known failure modes.	Do not alter identity; no AI glow, waxy skin, warped hands, fake text, unreadable logos, floating subject, or mismatched shadows.
10. QC instruction	Force final self-check.	Before final output, verify face preservation, skin tone, hands, shadows, perspective, fabric physics, and overprocessing.

Preservation-First Editing Rules

- Place identity constraints before creative instructions.
- When a person is in the image, never regenerate the face unless the user asks.

- Do not “beautify” by changing facial structure, nose, lips, jaw, eyes, or complexion.
- If skin tone adjustment is requested, base it on the reference image and keep it subtle.
- When changing clothing, keep the same person, pose, body proportions, lighting relationship, and camera angle.
- When changing background, match horizon line, camera height, lens perspective, light direction, and subject scale.

Positive Realism Keyword Library - Updated

Use case	Positive terms to use	Avoid overusing
Everyday realism	real photograph, natural light, candid, unposed, ordinary environment, imperfect texture, realistic shadows, believable perspective	masterpiece, cinematic perfection, ultra-polished, hyperreal
Phone-camera realism	modern smartphone photo, slight wide-angle perspective, computational HDR, mild edge sharpening, natural skin texture, realistic low-light noise	full-frame 85mm bokeh unless intended
35mm film realism	35mm film photograph, Kodak Portra 400, slight grain, natural halation, imperfect exposure, real lens softness	heavy fake grain everywhere
Fashion/editorial realism	real fashion editorial photo, garment drape, fabric weight, seams, stitching, restrained colour grade, natural skin texture	plastic runway render, fake logo, waxy skin
Product/e-commerce realism	sharp product photo, accurate colour, visible fabric weave, zipper teeth, stitching detail, controlled studio lighting, clean background	dramatic film look that shifts product colour
Street realism	real street photo, camera-height perspective, local signage, curb/road logic, environmental imperfections, real pedestrian scale	surreal empty city, impossible roads, generic fantasy city

Negative Constraint Library - Targeted, Not Universal

Failure type	Negative constraints
Identity drift	Do not change the face, facial proportions, skin complexion, age, ethnicity, hairstyle, expression, body structure, or recognisable identity.
AI glow/plastic look	No AI glow, waxy skin, porcelain texture, over-smoothing, overdone HDR, fake bloom, hyper-

	saturated colours, plastic lighting.
Anatomy failure	No warped hands, extra fingers, fused joints, distorted limbs, broken wrists, unrealistic posture, or melted facial features.
Text/logo failure	No pseudo-text, no fake logos, no misspelled brand marks, no unreadable signage. Avoid adding new text unless explicitly required.
Background mismatch	No mismatched lighting, floating subject, wrong horizon line, impossible perspective, inconsistent shadows, or synthetic background depth.
Fabric/clothing failure	No melted fabric, rubber-like cloth, missing seams, warped zippers, distorted logos, fake stitching, or clothing that ignores body shape.
Overprocessing	No excessive sharpening, crunchy skin, heavy grain overlay, excessive blur, crushed detail, or artificial vignette.

Prompt Modules for Common User Requests

Module 1 - Enhance This Picture Without Changing the Subject

Edit the provided image to improve realism and photo quality while preserving the subject exactly. Preserve the face, identity, age, complexion, body structure, hairstyle, expression, pose, clothing, and background unless the user specifically requested a change. Improve only exposure, natural contrast, white balance, realistic skin texture, fabric detail, contact shadows, lens realism, and subtle photographic imperfections. Avoid plastic skin, AI glow, altered facial features, changed skin tone, warped hands, fake bokeh, oversharpening, and over-smoothing. Before final output, verify the subject still looks like the same person.

Module 2 - Make This More Realistic and Less AI-Generated

Reduce the artificial AI-generated look by correcting physics first: perspective, contact shadows, light direction, material texture, object boundaries, and depth of field. Add only subtle real-camera traits: mild sensor/film grain, realistic lens softness, very slight chromatic aberration at high-contrast edges, natural colour grade, and believable ambient occlusion. Do not add heavy fake grain, harsh blur, overdone HDR, or random imperfections. Preserve all subject identity and scene content unless correction is required for realism.

Module 3 - Realistic Portrait Enhancement

Enhance the portrait as a believable real photograph. Preserve facial identity, natural complexion, pores, under-eye texture, beard/hairline edges, eye catchlights, and natural facial asymmetry. Improve exposure and white balance without changing face shape or skin tone. Keep skin realistic, not waxy or airbrushed. Correct hands and clothing only if they are visibly distorted. Use restrained Lightroom-style editing.

Module 4 - Clothing Change to Saint Laurent-Style Luxury Outfit

Change only the clothing to a realistic Saint Laurent-style luxury outfit appropriate to the pose, lighting,

and body shape. Preserve the person identity, face, complexion, body proportions, hands, pose, and expression. Make the garment physically believable: correct fabric weight, drape, seams, collar, sleeves, folds, stitching, and shadows. Avoid fake or unreadable logos. If a logo is needed, keep it minimal and clean; otherwise imply the brand through cut, tailoring, silhouette, material, and styling rather than large text.

Module 5 - Ottawa / Rideau Street Spring Background

Replace the background with a believable Rideau Street, Ottawa, Canada street environment at the start of spring, April 23, 2026. Match the subject camera height, horizon line, lens perspective, scale, light direction, and weather. Include realistic Ottawa urban details: sidewalks, curb height, road lanes, storefront rhythm, early spring trees/buds, no heavy snow, possible damp pavement, light jackets, and everyday pedestrian/traffic context. Do not place Parliament directly on Rideau Street if it would be geographically implausible. Keep the subject grounded with contact shadows and matching colour temperature.

Module 6 - Melato.ca Product Photo Enhancement

Enhance this apparel product image for a premium e-commerce page on melato.ca. Preserve the exact product design, colourway, logos, cut, fabric, and proportions. Improve sharpness, fabric weave, velour texture, zipper teeth, stitching, seams, cuffs, hems, and garment drape. Keep colour accurate in sRGB for web. Use clean premium lighting with natural shadows. Avoid changing the product design, adding fake details, oversaturating colour, plastic fabric, or distorted typography.

Module 7 - Natural Group Collage Without Altering Faces

Create a realistic group composition from the provided subjects while preserving every face and recognisable appearance exactly. Cut subjects cleanly from their backgrounds and place them into one coherent scene with matching scale, camera angle, lens perspective, light direction, shadow softness, grain/noise, colour temperature, and depth. Blend edges naturally without changing facial features, skin tones, body shapes, or expressions. Add contact shadows and shared ambient light so the group appears photographed together.

Camera, Lens, and Film Prompting - Updated Use

Camera and film references are useful realism anchors, but they must fit the image context. Do not force a Nikon FM2/Kodak Portra look onto every phone screenshot or e-commerce product image.

Scenario	Better camera language	Why
Everyday phone image	modern smartphone photo, 24-28mm equivalent, computational HDR, mild edge sharpening, deep focus	Matches common social-photo capture.
Street/fashion portrait	35mm or 50mm lens, natural window/street light, restrained editorial grade	Balances subject and environment.
Classic portrait	85mm lens, f/1.8, shallow depth of field, real catchlights	Good for controlled portrait; avoid if background swap needs

		context.
Product photo	studio product photograph, controlled softbox lighting, accurate colour, f/8-f/11 depth, macro detail	Product needs clarity more than cinematic blur.
Night street	available-light street photo, mixed colour temperatures, realistic highlight bloom, mild high-ISO noise	More believable than perfect clean night render.
35mm film style	Kodak Portra 400, slight grain, mild halation, imperfect exposure, natural colour	Use when user wants film aesthetic, not for colour-critical product.

Scenario Formulas

Scenario	Formula
Portrait enhancement	[Task] + [Preserve identity] + [skin/hair/eyes realism] + [natural light] + [subtle Lightroom grade] + [negative identity constraints] + [QC].
Street realism	[Subject/action] + [real location details] + [camera height/lens] + [weather/season] + [shadows/reflections] + [avoid impossible architecture/text] + [QC].
Fashion outfit swap	[Change clothing only] + [preserve body/face/pose] + [garment construction] + [brand styling if allowed] + [lighting/folds/shadows] + [no fake logos] + [QC].
Product image	[Preserve product] + [accurate colour] + [fabric/zipper/stitching] + [studio lighting] + [web output] + [no design changes] + [QC].
Group collage	[Use provided subjects] + [face lock] + [scale/perspective match] + [shared light/shadow/grain] + [natural edges] + [no identity drift] + [QC].
AI-artifact repair	[Identify exact artifact] + [repair only that area] + [preserve all else] + [physics/material correction] + [targeted negatives] + [QC].

Repair Loop for Gemini / Nano Banana

1. Inspect the result for identity drift, skin tone shift, hand distortion, material errors, text/logo failures, shadow mismatch, background mismatch, and overprocessing.
2. Classify the failure: identity, anatomy, lighting, perspective, texture, text/logo, background, colour, or compression.
3. Rewrite a targeted correction prompt. Do not regenerate the entire image unless the failure is global.
4. Restate preservation constraints before the correction.

5. Apply the smallest necessary fix.
6. Stop when the image is believable; do not keep adding realism effects after the core problem is solved.

Quality-Control Checklist

- Face still matches original subject?
- Skin complexion preserved or adjusted only as requested?
- Hands, fingers, wrists, and posture physically plausible?
- Clothing follows body shape, gravity, seams, folds, and fabric weight?
- Text/logos avoided or readable and stable?
- Subject grounded with contact shadows and ambient occlusion?
- Background matches camera height, horizon, perspective, season, and lighting?
- Depth of field consistent with focus plane and camera scenario?
- No excessive grain, sharpening, blur, HDR, vignette, or AI glow?
- Final result looks like a photograph, not a render or filter stack?

Quick Reference Summary

- Prompt order: task -> preserve -> target change -> scene -> camera -> lighting -> texture -> style -> negatives -> QC.
- Preserve identity before creative editing.
- Use targeted negatives, not universal negative dumps.
- Physics and integration matter more than camera-name decoration.
- For product images, preserve design and colour accuracy before stylisation.
- For background changes, match camera height, horizon line, light direction, season, and scale.

Source Notes and Reliability Labels

This document gives operational image-analysis and image-editing guidance. It must not claim legal or forensic certainty without original files, metadata, provenance, and human review. Use probability language.

- **Internal AuthenticLens Document 1 v1.0:** Source baseline for the AI artifact lexicon: anatomical anomalies, text failures, pattern distortions, boundary failures, AI glow, material physics, structural coherence, and model signatures.
- **Internal AuthenticLens Document 2 v1.0:** Source baseline for photographic physics: optical aberrations, vignetting, distortion, flare, diffraction, sensor noise, film grain, depth of field, reflection physics, exposure, tone, and colour temperature.
- **Internal AuthenticLens Document 3 v1.0:** Source baseline for realism prompting: positive/negative realism vocabulary, film stock and camera references, scenario prompt formulas, and platform strategies.
- **Google AI Developers - Image generation with Gemini:** Current source note: Gemini supports conversational image generation/editing with text, images, or both. Nano Banana refers to Gemini 2.5 Flash Image using model gemini-2.5-flash-image. Nano Banana Pro refers to Gemini 3 Pro Image Preview

using model gemini-3-pro-image-preview. Google states generated images include SynthID watermarking.

- **Adobe Lightroom Help - Effects and Details:** Current source note: Texture can smooth or accentuate texture without changing colour/tonality; Clarity changes edge contrast; Dehaze changes haze and can affect perceived detail/colour; Vignette and Grain are stylistic effects; Detail controls Sharpening, Noise Reduction, and Color Noise Reduction.
- **C2PA Technical Specification:** Current source note: Content Credentials use signed manifests and assertions to record provenance and edits. Absence of C2PA data is not proof that an image is fake, and presence of metadata should be interpreted with context.
- **W3C / IEC sRGB references:** Current source note: sRGB is the default web colour space; for e-commerce and social delivery, unmanaged or non-sRGB exports can shift colour appearance across devices.
- **IPTC Photo Metadata Standard:** Current source note: IPTC metadata structures descriptive, location, rights, and administrative data for professional image workflows. Metadata is useful context but is not standalone authenticity proof.

Reliability labels used inside the document: Confirmed source, Common photography practice, Image-forensics heuristic, Operational guidance, Prompting heuristic, Model-behaviour observation.